

SLOWLY-ROTATING TERRESTRIAL ATMOSPHERES

FROM EARTH TO VENUS

Greg Colyer & Geoffrey K. Vallis

University of Exeter

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Venus

1. Rotation period ~ 200 Earth days.
2. Surface pressure 92 bars
3. Largely CO_2 atmosphere
4. Surface temperature ~ 700 K
5. Wide Hadley Cell, extends to $\sim 60^\circ$.
6. Super-rotates: winds $\sim 100 \text{ m s}^{-1}$ at equator at cloud level

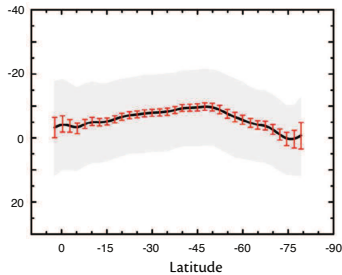
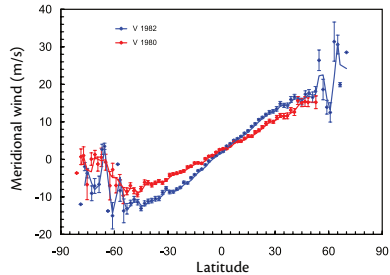
Today's topics

1. Effects of rotation and depth.
2. Hadley Cell
 - The classical theory re-visited.
3. Multiple equilibria in Venusian atmospheres.

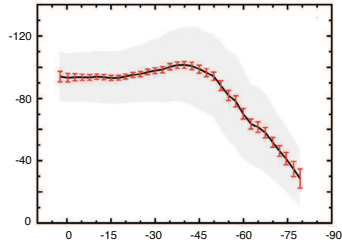
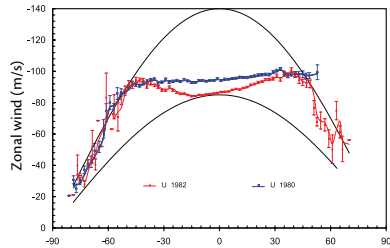


Dayside only

Venus Express orbiter.



Meridional wind



Zonal wind

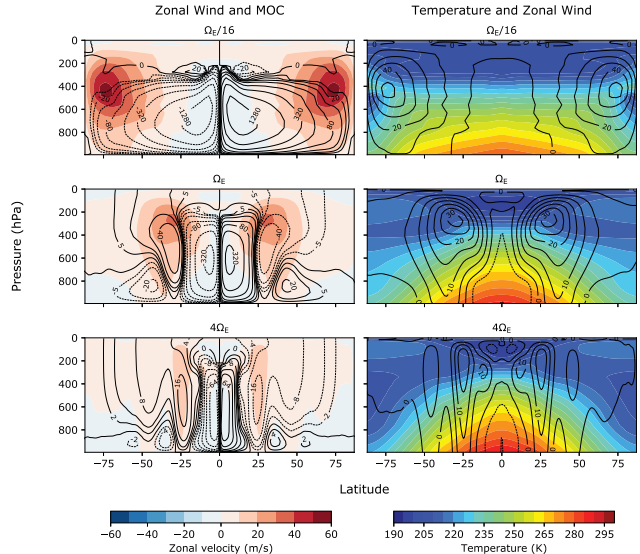
Limaye (2007),
Khatuntsev et al (2013)

HADLEY CELL FOR LOW ROTATION



- How far does the Hadley Cell go?
- What values does zonal wind take?
- Then what happens at zero rotation?
- Venus GCMs are all over the place and are very sensitive to parameter choices.

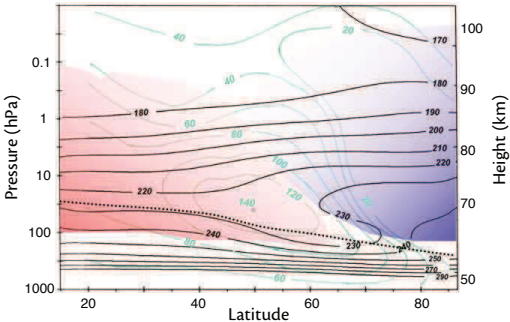
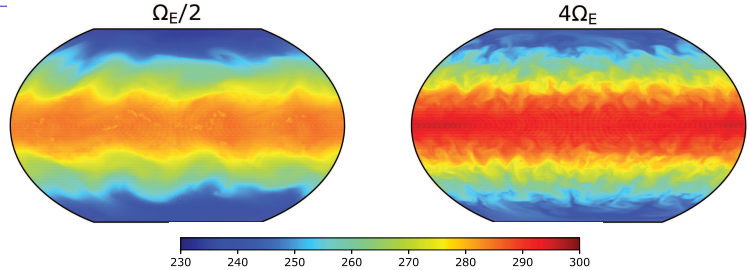
(Simulations by J. Eager with Isca.)



NEAR SURFACE TEMPERATURE AS ROTATION CHANGES



GCM Simulation:



Venus observation
(Titov, 2013, orbiting infra-red
radiometer)

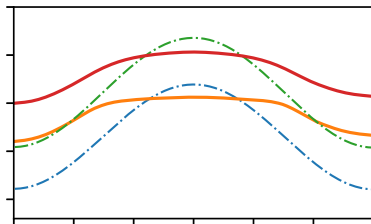
Meridional temperature gradient
evidently smaller at low rotation.

TEMPERATURE AND VELOCITY, $\Omega = \Omega_E/5$

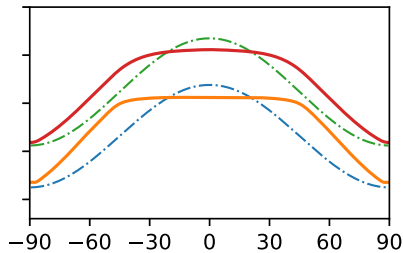


'3D' simulation

Temperature
 $\Omega_E/5$ ($m \leq 2$ only), T170

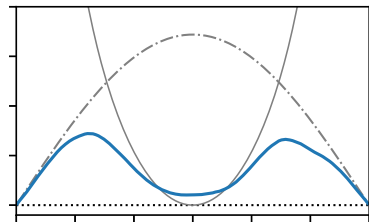


$\Omega_E/5$ ($m = 0$ only), T170

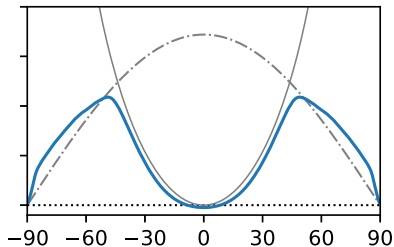


Axi-symmetric

Zonal wind
 $\Omega_E/5$ ($m \leq 2$ only), T170



$\Omega_E/5$ ($m = 0$ only), T170

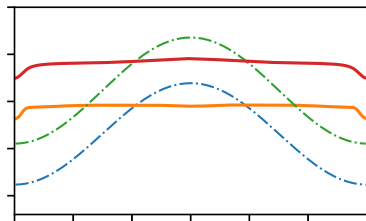


TEMPERATURE AND VELOCITY, $\Omega = \Omega_E/100$

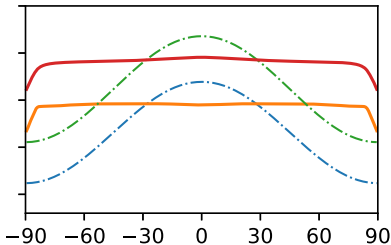


'3D' simulation

Temperature
 $\Omega_E/100$ ($m \leq 2$ only), T170

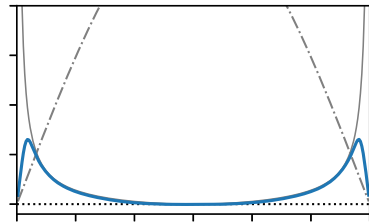


$\Omega_E/100$ ($m = 0$ only), T170

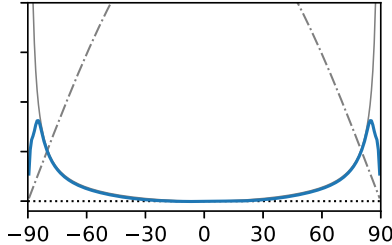


Axi-symmetric

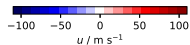
Zonal Wind
 $\Omega_E/100$ ($m \leq 2$ only), T170



$\Omega_E/100$ ($m = 0$ only), T170



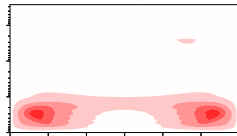
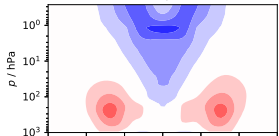
ZONAL MEAN ZONAL WIND, SIMULATIONS



(a) $\Omega = \Omega_E, p_s = 1 \text{ bar}$

(b) $\Omega = \Omega_E/20, p_s = 1 \text{ bar}$

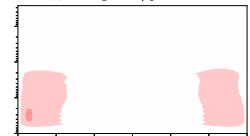
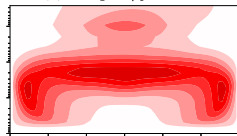
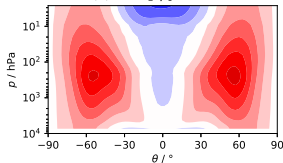
(c) $\Omega = \Omega_E/100, p_s = 1 \text{ bar}$



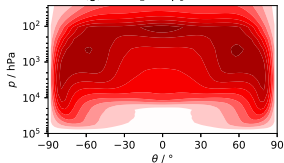
(d) $\Omega = \Omega_E, p_s = 7.9 \text{ bar}$

(e) $\Omega = \Omega_E/20, p_s = 7.9 \text{ bar}$

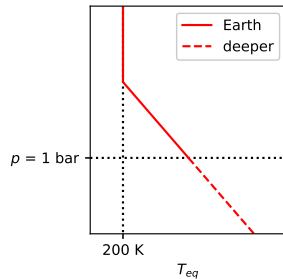
(f) $\Omega = \Omega_E/100, p_s = 7.9 \text{ bar}$



(g) $\Omega = \Omega_E/20, p_s = 92 \text{ bar}$



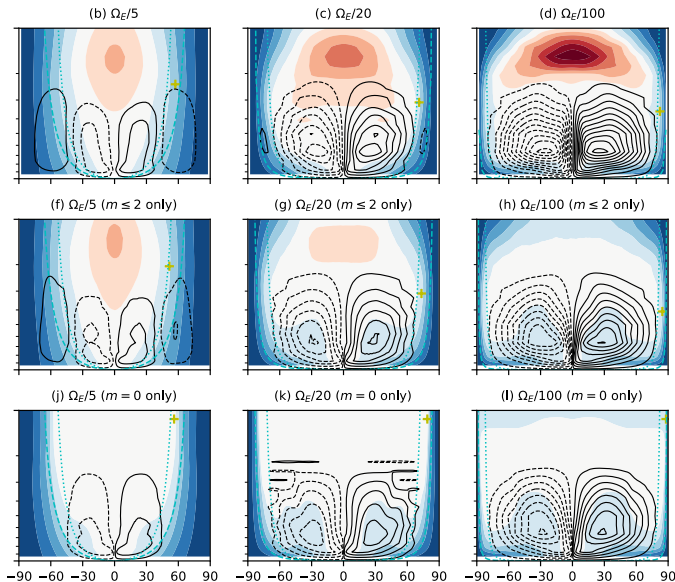
- (i) As Ω decreases, Hadley cell widens; zonal winds strengthen, then, at very low rotation rates, weakens.
- (ii) As p_s increases: superrotating layer strengthens and deepens (between Hadley cell and tropopause)



3D primitive equations.
T42, 30 level simulations,
with Isca.

SUPER-ROTATION

Spontaneously appears at low rotation



Earth-depth ($p_s = 1 \text{ bar}$)

Colors are:

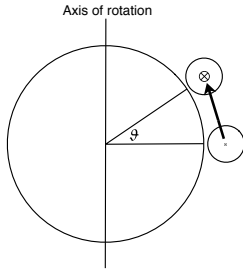
$$M/\Omega a^2 - 1$$

where

$$M = (u + \Omega a \cos \theta) a \cos \theta$$



(E.K. Schneider, Held and Hou)



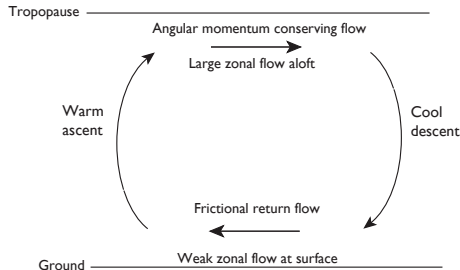
Assume flow is axi-symmetric.
Outflow is angular momentum conserving:

$$U = \Omega a \frac{\sin^2 \theta}{\cos \theta}$$

Temperature from thermal wind balance:

$$T = T(0) - \frac{T_0 \Omega^2 a^2 \theta^4}{2gH}$$

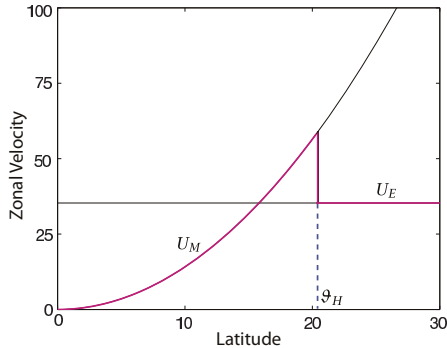
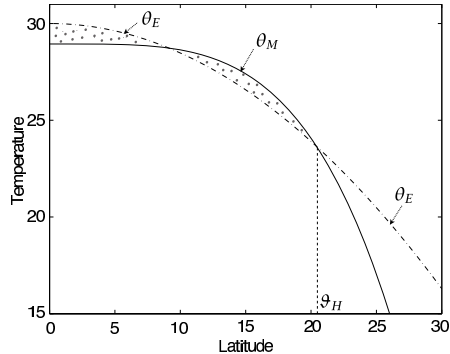
Temperature falls with latitude, gets too cold and sinks.



HADLEY CELL CALCULATIONS



Original theory



θ_E = radiative equilibrium temperature,

θ_M = temperature from angular momentum conserving wind (via thermal wind). Discontinuity in zonal wind at edge of Hadley Cell.

Transition latitude:

$$\theta_H = \left(\frac{5\Delta\theta_h g H}{3a^2 \Omega^2 \theta_0} \right)^{1/2} \propto \frac{1}{\Omega}.$$

Goes further poleward as rotation falls.



Skip this slide (!)

Steady zonal momentum equation with $w = 0$ and no eddies.

$$v \left(\frac{1}{a} \frac{\partial u}{\partial \theta} - f - \frac{u \tan \theta}{a} \right) = 0$$

Either:

- (i) $v = 0$ (identically). Radiative equilibrium solution.
- (ii) Or, angular momentum conserving solution:

$$\frac{1}{a} \frac{\partial u}{\partial \theta} - f - \frac{u \tan \theta}{a} = \frac{1}{a^2 \cos \theta} \frac{\partial M}{\partial \theta} = 0$$

So that: $M = (u + \Omega a \cos \theta) a \cos \theta = \text{constant}$

$$u = u_M(\theta) + \frac{u_0}{\cos \theta} = \frac{\Omega a \sin^2 \theta}{\cos \theta} + \frac{u_0}{\cos \theta}$$

- Low latitudes – momentum-conserving solution
- High latitudes – no circulation (radiative equilibrium).



Skip this slide (!)

$$fu + \frac{u^2 \tan \theta}{a} = -\frac{gH}{a\Theta_0} \frac{\partial \bar{\Theta}}{\partial \theta}$$

Angular Momentum Conserving Regime:

With $u = u_M$ and $u_0 = 0$ solution is

$$\frac{\bar{\Theta}(0) - \bar{\Theta}}{\Theta_0} = \frac{u_M^2}{2gH} = \frac{\Omega^2 a^2 \sin^4 \theta}{2gH \cos^2 \theta}. \quad (1)$$

Radiative Equilibrium Regime:

Given a typical $\Theta_E(\theta)$ the solution is

$$u_E = \Omega a \cos \theta \left(\sqrt{2R + 1} - 1 \right), \quad \text{where} \quad R \equiv \Delta_H g H / \Omega^2 a^2. \quad (2)$$

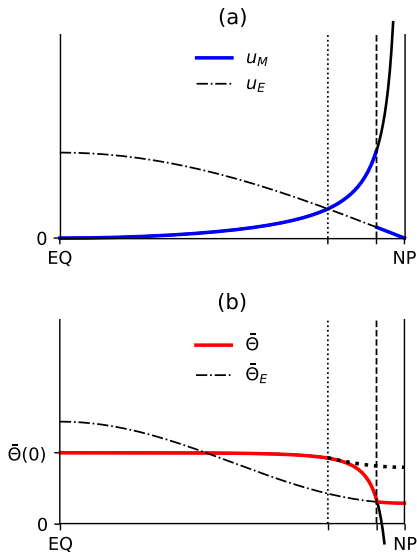
For small Ω :

$$u_E = \sqrt{2\Delta_H g H} \cos \theta \approx \sqrt{2\Delta_H g H} \varphi \quad (3)$$

where φ is co-latitude. u_E is independent of Ω .



Velocity discontinuity is more pronounced



1. Continuity of temperature at θ_H :

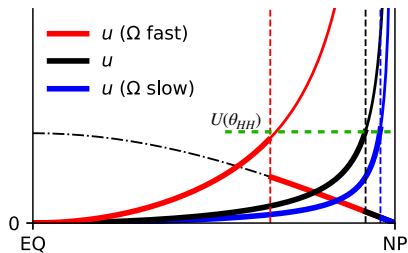
$$\bar{\theta}(\theta_H-) = \bar{\theta}_E(\theta_H+)$$

2. Closure of the energy budget over the Hadley cell $\theta < \theta_H$:

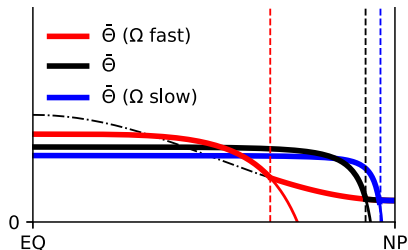
$$\int_0^{\theta_H} \bar{\theta} \cos \theta d\theta = \int_0^{\theta_H} \bar{\theta}_E \cos \theta d\theta.$$

(Equal area construction).

EFFECTS OF ROTATION RATE



(b)



Low-rotation limit, $\Omega \rightarrow 0$:

$$\frac{\bar{\theta}(0) - \bar{\theta}}{\Theta_0} = \frac{\Omega^2 a^2 \sin^4 \theta}{2gH \cos^2 \theta} \rightarrow 0 \quad \text{unless } \theta \rightarrow \pi/2.$$

Let $\varphi \equiv \pi/2 - \theta \ll 1$:

$$\frac{\bar{\theta}(0) - \bar{\theta}}{\Theta_0} = \frac{\Omega^2 a^2}{2gH \varphi^2}$$

Transition latitude:

$$\varphi_H = \frac{\sqrt{3}\Omega a}{2\sqrt{\Delta_H g H}} \propto \Omega$$

Note that velocity at transition latitude does not depend on rotation rate!



Near pole, angular momentum conserving wind is:

$$u_M(\varphi) = \frac{\Omega a}{\varphi}$$

Wind at the vortex edge (in original theory):

$$u(\varphi_H) = \frac{\Omega a}{\varphi_H} = \frac{2\sqrt{\Delta_H g H}}{\sqrt{3}}$$

cf Covey et al (1986)

– independent of Ω !

On the poleward side, gradient thermal wind balance:

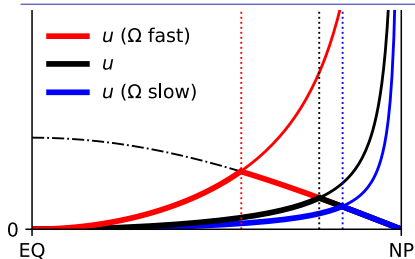
$$u_E(\varphi) = \sqrt{2\Delta_H g H} \varphi$$

At the edge of the polar vortex:

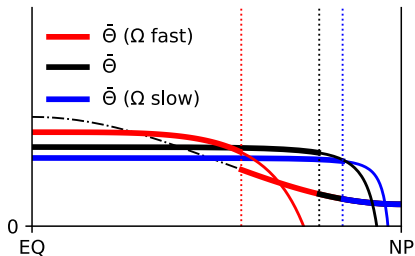
$$u_E(\varphi_H) = \sqrt{2\Delta_H g H} \varphi_H = \sqrt{\frac{3}{2}} \Omega a, \quad \frac{u_E}{\varphi a} = \frac{\sqrt{2\Delta_H g H}}{a}$$

Polar vortex rotates like a rigid body with relative angular velocity independent of Ω .

ALTERNATIVE MATCHING CONDITION: CONTINUOUS VELOCITY



(b)



- Let u be continuous at the edge of the Hadley Cell:

$$u_M(\theta_H-) = u_E(\theta_H+)$$

Boundary co-latitude now given by:

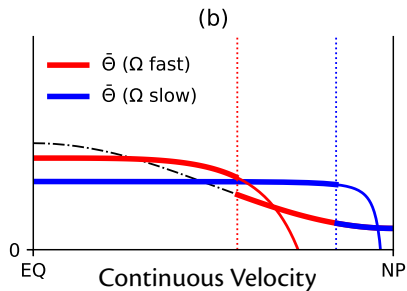
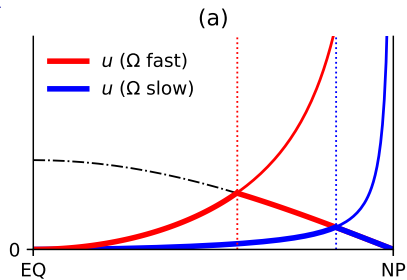
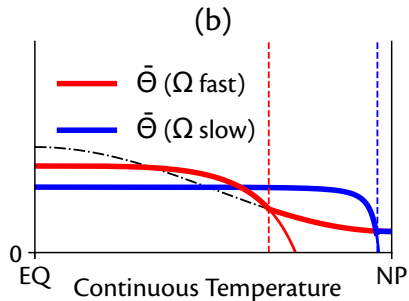
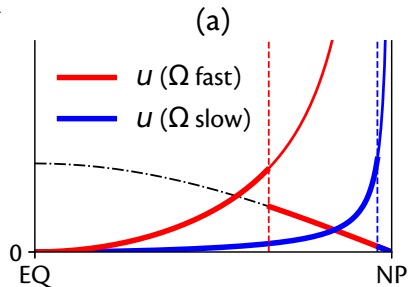
$$\varphi_H = \left(\frac{1}{2R} \right)^{\frac{1}{4}} = \frac{\sqrt{\Omega a}}{(2\Delta_H g H)^{\frac{1}{4}}}$$

- Maximum velocity now *does* fall with decreasing rotation.

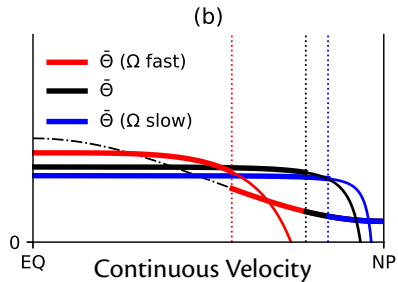
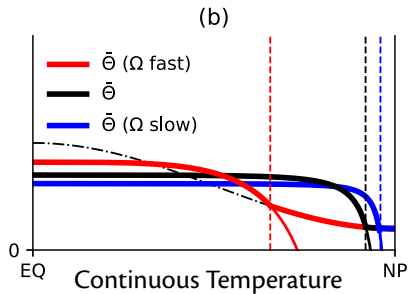
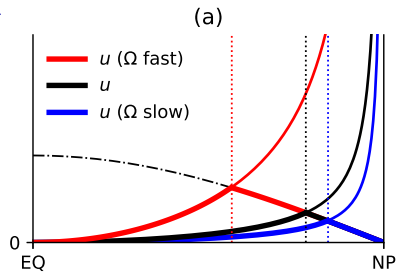
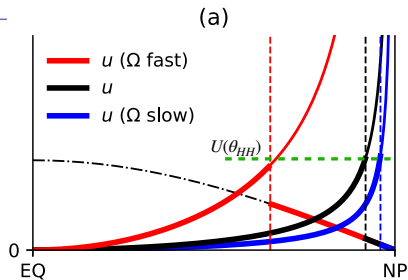
$$u(\varphi_{CV}) = \sqrt{\Omega a} (2\Delta_H g H)^{\frac{1}{4}}$$

- Discontinuity in temperature, $\bar{\theta}$, if thermal wind satisfied.

THEORY WITH DIFFERENT MATCHING CONDITIONS



THEORY WITH DIFFERENT MATCHING CONDITIONS



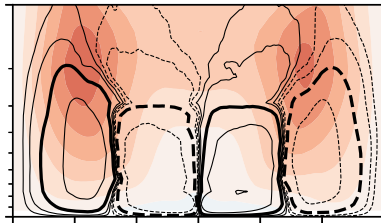
NON-ZERO MOC AT ALL LATITUDES

Keeps everything continuous

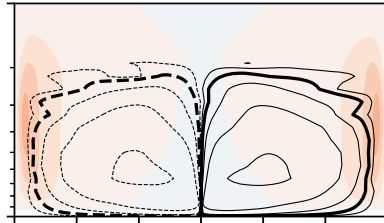


'3D' simulation

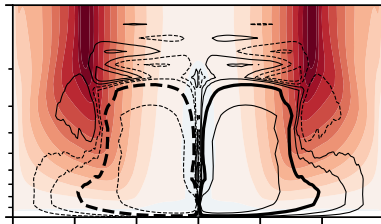
$\Omega_E/5$ ($m \leq 2$ only), T170



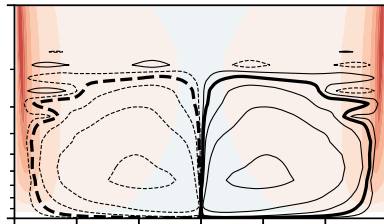
$\Omega_E/100$ ($m \leq 2$ only), T170



$\Omega_E/5$ ($m = 0$ only), T170



$\Omega_E/100$ ($m = 0$ only), T170



Axi-symmetric

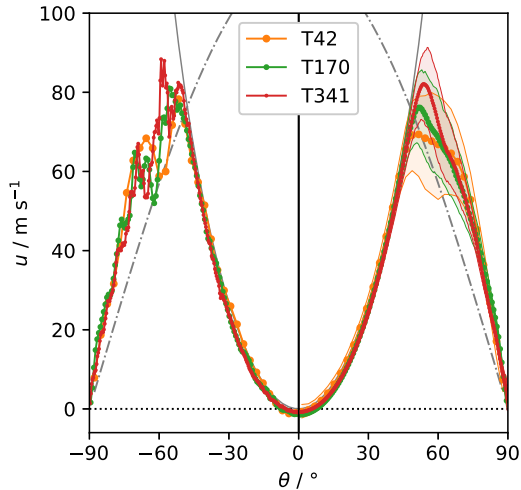
Weak direct Cell at
high latitudes

-90 -60 -30 0 30 60 90

-90 -60 -30 0 30 60 90

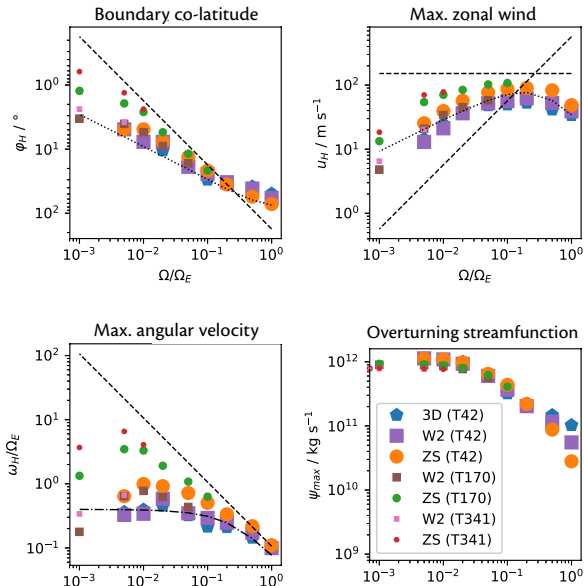
ZONAL WIND

Varies rapidly but is continuous



Zonal wind continuous.

Note that the wind is not equal to the radiative equilibrium wind at high latitudes.

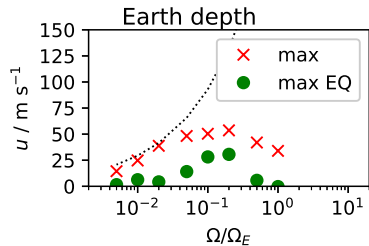




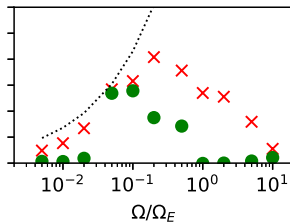
1. The simulations collapse at low rotation rates corresponding to Venus.
2. Simulations, although super-rotating, have velocities of a few tens of meters/sec, not 100 m/s.



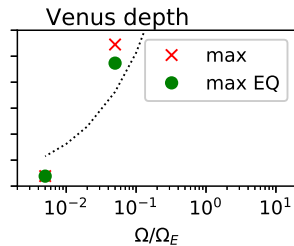
$p_s = 1$ bar



$p_s = 8$ bar



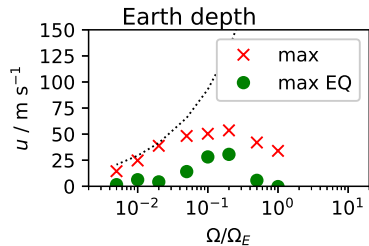
$p_s = 92$ bar



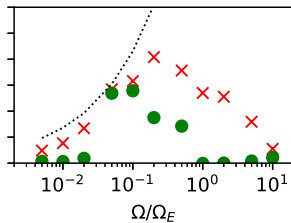
Although super-rotating, solutions collapse at very low rotation rates, even at Venus depth.



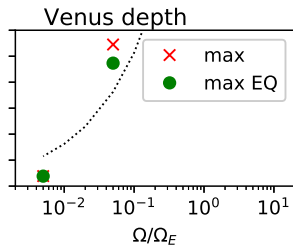
$p_s = 1$ bar



$p_s = 8$ bar



$p_s = 92$ bar



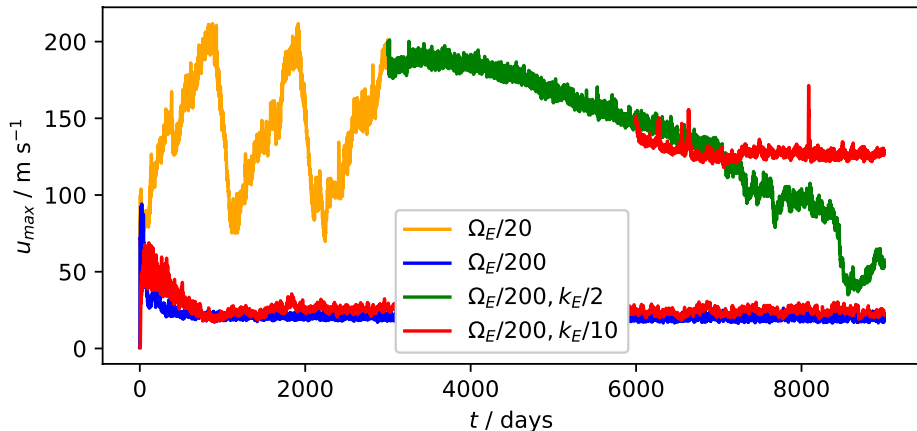
Although super-rotating, solutions collapse at very low rotation rates, even at Venus depth.

But wait!

MULTIPLE STATES IN VENUS-LIKE ATMOSPHERES

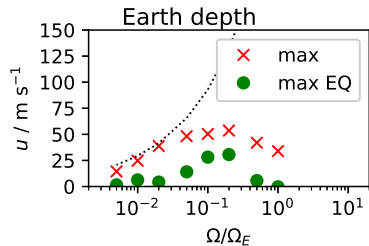


Different states appear with different initial conditions, provided the damping is small (radiative relaxation timescales are long).

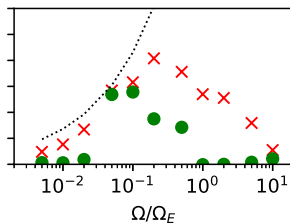




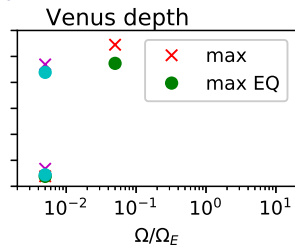
$p_s = 1$ bar



$p_s = 8$ bar



$p_s = 92$ bar



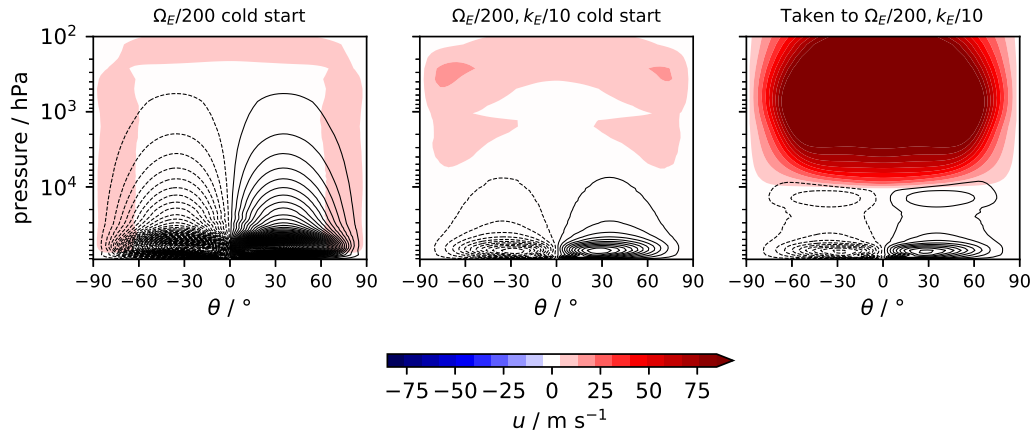
An additional solution exists at very low rotation rates at Venus depth.

Exactly same parameters, different initial conditions.

MULTIPLE STATES IN VENUS-LIKE ATMOSPHERES



Different slowly-rotating states.



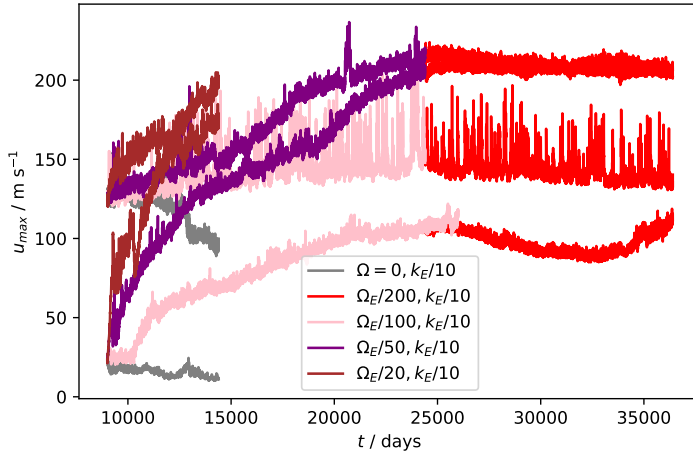
Right two plots have same parameters.

Note second layer of circulation, cf. Lebonnois (2016)

MULTIPLE STATES IN VENUS-LIKE ATMOSPHERES



Three possible solutions? More?



- Different initial conditions and pathways.
- Superrotation is long-lived even if Ω is set to zero (grey).
- A possible jet-like state at zero rotation? (grey lines)



- At low rotation rates Hadley Cell extends to high latitudes with nearly constant temperature.
- Superrotation is a generic feature at low rotation.
Well known. Interaction of Kelvin and Rossby waves. (Mitchell, Vallis, Potter etc.)
- Most simulations (and original theory) show maximum zonal wind falling at low rotation rates, collapsing almost completely at Venusian rotation rates.
- But there exist additional solutions at low rotation rates in deep atmospheres.
Can maintain high super-rotation at very low (Venusian) rotation rates.
- Exoplanets with slow rotation may not all behave like Venus.

The End



- In general the simulations fall roughly between the two sets of scalings. The boundary latitude follows HH quite well for ZS, provided sufficient resolution is used.
- In all simulations the maximum zonal wind does decrease as rotation rate decreases, but it is sustained at a higher level for ZS (highest resolution: convergence uncertain)
- Very interesting that W2 well converged and much closer to continuous- u theory.
- Neither ZS theory could apply exactly to a GCM simulation, in which one would not expect actual discontinuities to be maintained!
- Note HH 'paradox' that maximum u stays constant even as $\Omega \rightarrow 0$. In the continuous- u theory, the polar vorticity stays constant, but because the polar vortex shrinks, $u \rightarrow 0$ everywhere (but there is still a nonzero overturning circulation: hollow points).